Programming Course C++

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Exercise Sheet 1

- These exercises should be done in groups of two.
- After completing the exercises, create a tar-file using e.g. the command

  `tar -zcvf Exercise01_yourgroup.tgz <list of files>`

  and then upload the resulting file in Stud.IP as shown in the lecture.
- The exercises marked with ★ should be reviewed for the written exam.
- Mind the deadline for this week's exercise sheet (see Stud.IP for details).

What you should know after this exercise:

- What variables are and how are they used in a C++-program.
- What the terms `stdout`, `stderr`, and `stdin` mean and how you can redirect output to a file.
- How compiler messages can be used to find errors in your code.

Problem 1.1: Sphere *(7 points)*

Write a program `Sphere.cpp` that asks the user to enter a value for the radius of a sphere and then prints its volume and surface area with four significant digits. The output of the program should like this:

```
Please enter a value for radius: 1.5
The sphere with radius 1.5 has a volumne of 14.14 and a surface area of 28.27
```

Optional question: how can you change the output to a fixed number of decimal places? Include the file `Sphere.cpp` in your uploaded solution.

★ Problem 1.2: Basic I/O-Redirection *(4 points)*

Download, compile and run the program `Output.cpp`. The output should be this:

```
This is written to stdout
This is written to stderr
```

Write down the commands to achieve the following:

a) `stdout` is written to a file `out.log` (this file should contain the first line of output only).

b) `stderr` is written to a file `err.log` (this file should contain the second line of output only).

c) `stdout` and `stderr` are written to a single file `both.log`.

Inlcude a text file with the answer (i.e. the commands you have used) in your uploaded solution.

★ Problem 1.3: Compiler Errors *(5 points)*

Download the program `Divide.cpp` from Stud.IP and compile it. You will notice a lot of error messages from the compiler. Correct the program and answer the following questions:

a) Identify the errors in the program code and the corresponding error message from the compiler. Write both into a separate textfile or in the comments of the corrected code (if the compiler error is very long you can shorten it to the important two or three lines).

b) Optional task: write down the command with I/O-redirection to store the compiler errors directly into a file (you may use this file to answer a)).

c) Test the program after it compiles correctly. Is the output as expected (try out entering different numbers)? If not, where is the problem?

d) The last lines of the code read

```
double y,z;
cout << "y = " << y << endl;
```

While this part of the code is formally correct it has the potential to cause problems. How can you modify the `g++` compiler command to get additional warnings? Also explain the warnings.

Do not forget the general hints that were given in the lecture. Your uploaded answer should include the modified and correctly compiling `Divide.cpp` and possibly an additional text file with your answers.

Problem 1.4: Addition Problems *(4 points)*

Download the program `Additons.cpp` from Stud.IP, compile and run it. Then look at the code and do the following:

a) Identify `C++`-statements that were not yet discussed in the lecture. Briefly explain the purpose of these statements (you can add a comment to the code or write a separate text file).

b) Modify the program so that it prints a message after each answer given. The messages should read

```
You gave the correct answer and get one point!
```

or

```
The correct answer is ..., you lose one point!
```

depending on the answer given (replace the `...` with the value of $a + b$, of course).

Your uploaded answer should include the modified `Additions.cpp` and possible an additional answer text file.